



Predicting UK Road Collision Severity
Using Data Mining Classification Techniques

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AI306 - Data Mining Course Project

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Abstract

This project develops a complete data mining workflow for predicting road collision severity using the UK Road Safety Collision 2023 dataset. The target variable is `collision_severity` with three classes: Fatal, Serious, and Slight. The dataset is highly imbalanced because Fatal collisions represent only 1.46% of all records, while Slight collisions represent more than 76%.

Several imbalance-handling methods were tested, including class weighting, random undersampling, random oversampling, and SMOTE. The project also included diagnostic experiments for data leakage and overfitting. A leakage diagnostic showed that including outcome-related columns produced unrealistically perfect results, proving that those columns allowed the model to cheat. These columns were removed from the final models.

The required models were implemented and compared: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The best final model was the Random Forest Class Weighted Regularized model with a macro F1-score of 0.3816. The results show that class imbalance handling and leakage prevention are very important for responsible road safety prediction.

1. Introduction

Road collisions are a major public safety concern because they can lead to injuries, fatalities, economic costs, and pressure on emergency and health services. This project studies whether road, environmental, and temporal conditions can be used to predict collision severity. The prediction task is formulated as supervised multi-class classification because each record has a known severity label: Fatal, Serious, or Slight.

The course project requires a complete data mining workflow including data exploration, preprocessing, model building, evaluation using appropriate metrics, interpretation of results, and final conclusions. This report follows that structure and highlights important modelling problems such as class imbalance, data leakage, misleading accuracy, and overfitting risk.

The main research question is:

Can the severity of a road collision be predicted using road, environmental, and temporal features available before or during the collision?

2. Literature Review

Road safety prediction is commonly studied as a classification problem, where historical collision records are used to learn patterns associated with different severity levels. Official road safety datasets such as STATS19 are valuable because they contain structured features describing collision conditions, road type, time, weather, lighting, and road surface.

A common challenge in collision severity prediction is class imbalance. Slight collisions are much more frequent than Fatal collisions. If this problem is ignored, a classifier can achieve high accuracy by predicting the majority class while failing to identify rare Fatal cases. For this reason, previous research recommends using recall, F1-score, macro averages, cost-sensitive learning, and resampling methods instead of accuracy alone.

SMOTE creates synthetic minority-class examples instead of simply duplicating existing rows. This can improve minority-class learning, although it must only be applied to the training set to avoid leakage. Random Forest is widely used for structured tabular data because it combines many decision trees and can provide feature importance. Gradient Boosting is another ensemble method that builds models sequentially to correct previous prediction errors.

3. Experiments

3.1 Data Description

The dataset used in this project is the UK Road Safety Collision 2023 file. Each row represents one reported road collision. The target variable is `collision_severity`. The collision file was selected because it contains one record per collision and directly includes the target variable. Casualty and vehicle files were not used as the main dataset because they record injured people and vehicles separately, which can duplicate collision records.

After filtering valid target classes, the final cleaned dataset contained 104,258 rows. The stratified train-test split produced 83,406 training rows and 20,852 testing rows.

Table 1. Target class distribution

Collision Severity	Label	Count	Percentage
1	Fatal	1522	1.4600
2	Serious	23438	22.4810
3	Slight	79298	76.0590

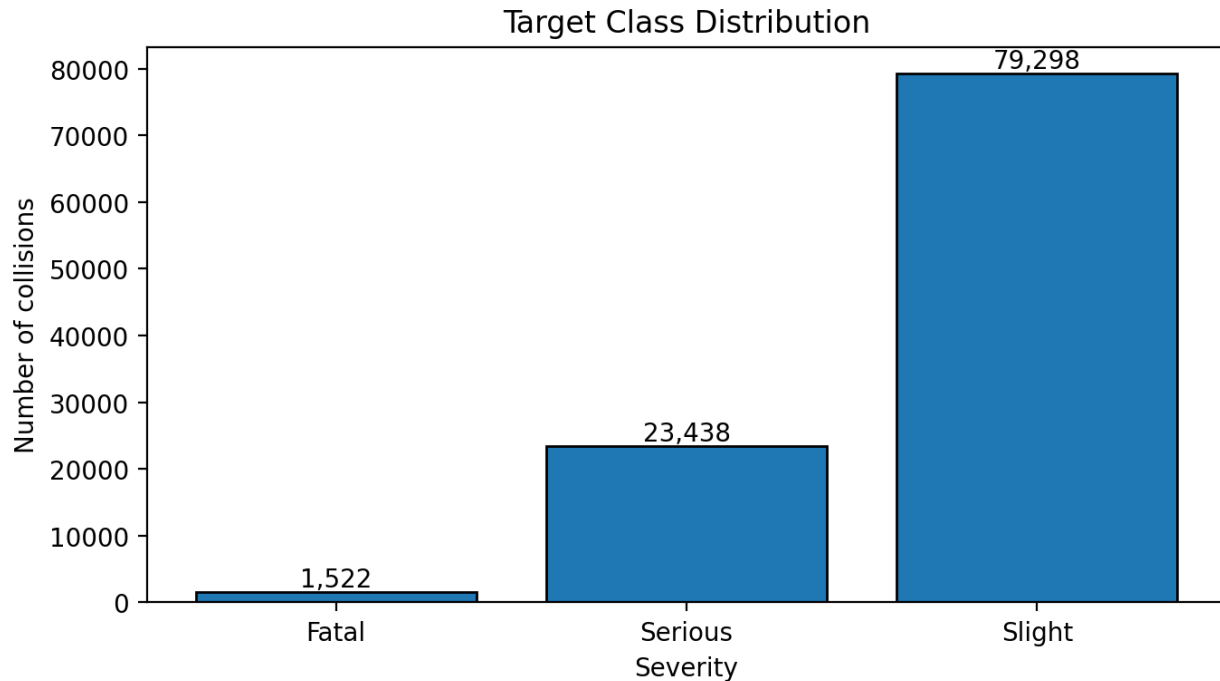


Figure 1. Collision severity distribution showing severe class imbalance.

3.2 Data Preprocessing

The preprocessing stage converted dates and times into useful features, handled missing and unknown values, removed leakage variables, and encoded categorical features. The engineered features were month, hour, is_weekend, and time_period.

The time_period feature used four intervals:

- Morning: 05:00-11:59
- Afternoon: 12:00-16:59
- Evening: 17:00-20:59
- Night: 21:00-04:59

Categorical coded -1 values were treated as "Unknown" instead of dropping rows. Numeric invalid values were treated as missing and handled using median imputation. Categorical variables were encoded using one-hot encoding with handle_unknown="ignore".

The train-test split was stratified using test_size=0.20 and random_state=42. No artificial row cap was used.

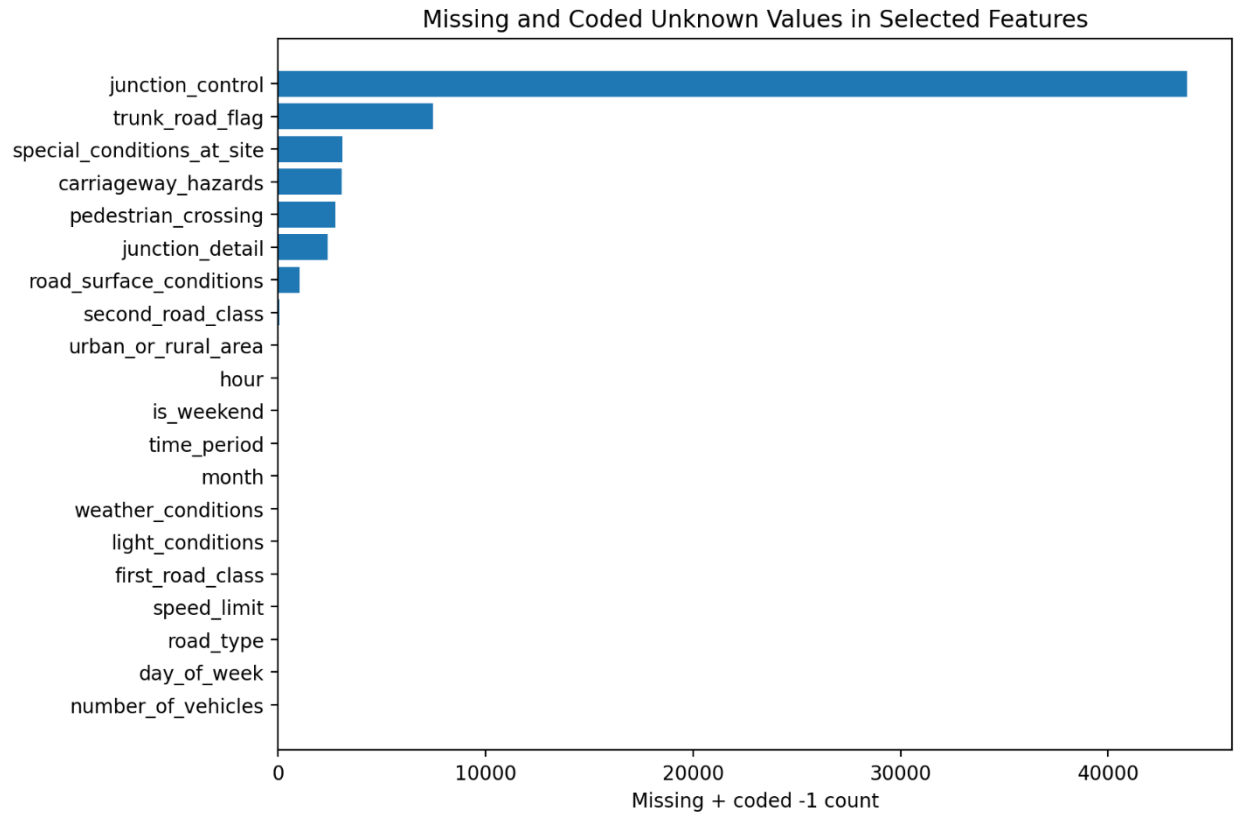


Figure 2. Missing and coded unknown values in selected features.

Table 2. Main project problems and implemented solutions

Problem Found	Evidence	Implemented Solution
Class imbalance	Fatal collisions were only 1.46% of records.	Compared no balancing, class weighting, random undersampling, random oversampling, and SMOTE; used macro F1 and minority recall.
Misleading accuracy	Majority baseline had high accuracy but zero Fatal and Serious recall.	Used macro F1, macro recall, Fatal recall, Serious recall, and confusion matrices.
Data leakage	Outcome-related columns produced 100% test accuracy in a diagnostic model.	Excluded leakage columns from final models.
Overfitting risk	Tree-based models can memorize patterns or leakage features.	Used leakage removal, max_depth, min_samples_leaf, max_features, and train-vs-test gap checks.
Unknown coded values	Some columns use -1 to represent unknown/not applicable.	Converted categorical -1 to Unknown and imputed numeric invalid values.
Gradient Boosting class weights	GradientBoostingClassifier has no class_weight parameter.	Used balanced sample weights during fitting.

3.3 Models Built

The required models were implemented and compared:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting

A Majority Baseline model was also included to demonstrate why accuracy alone is misleading for imbalanced datasets.

Logistic Regression used `class_weight="balanced"`. Decision Tree used SMOTE with regularization. Random Forest used `balanced_subsample` class weighting with regularization. Gradient Boosting used balanced sample weights because `GradientBoostingClassifier` does not directly support class weights.

3.4 Class Imbalance Experiments

The imbalance problem was solved through controlled experiments instead of only being reported. A regularized Decision Tree model was used to compare five imbalance-handling methods:

- No balancing
- Class weighting
- Random undersampling
- Random oversampling
- SMOTE

SMOTE achieved the best macro F1-score in the controlled imbalance experiment, while Random Oversampling achieved higher Fatal recall. Random undersampling improved Fatal recall but removed too much majority-class information, which reduced overall performance.

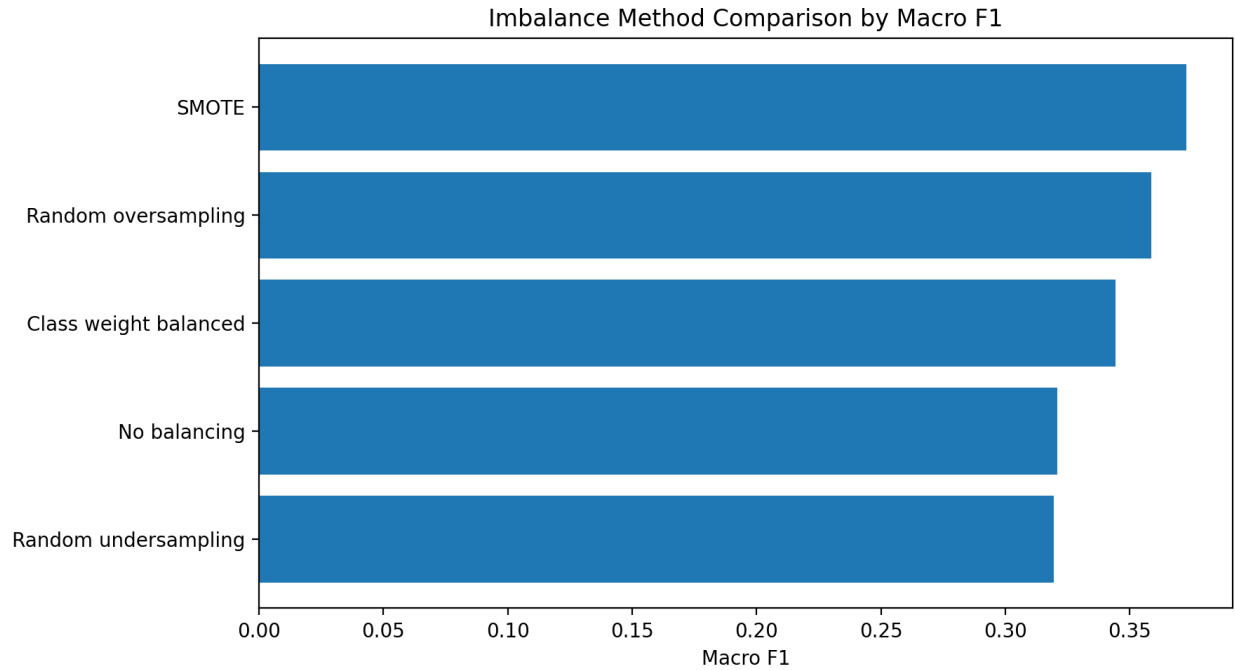


Figure 3. Comparison of imbalance handling methods using macro F1-score.

Table 3. Imbalance method comparison

Method	Accuracy	Macro F1	Weighted F1	Fatal Recall	Serious Recall	Slight Recall	Resampled Rows
SMOTE	0.6612	0.3727	0.6630	0.1645	0.2280	0.7987	190314
Random oversampling	0.5465	0.3587	0.5995	0.3059	0.3863	0.5985	190314
Class weighted	0.5087	0.3442	0.5725	0.3980	0.3752	0.5503	83406
No imbalance handling	0.7560	0.3210	0.6772	0.0000	0.0591	0.9765	83406
Random undersampling	0.4638	0.3194	0.5363	0.5066	0.3215	0.5051	3654

Table 4. Training class counts after imbalance methods

Method	Class 1 Count	Class 2 Count	Class 3 Count
No imbalance handling	1218	18750	63438
Class weighted	1218	18750	63438
Random undersampling	1218	1218	1218
Random oversampling	63438	63438	63438
SMOTE	63438	63438	63438

3.5 Final Model Results

The final model comparison used macro F1-score as the main evaluation metric because the dataset is highly imbalanced. Accuracy was still reported, but it was not used as the main selection criterion because high accuracy can still occur when minority classes are poorly predicted.

Five models were compared:

- Majority Baseline
- Logistic Regression Class Weighted
- Decision Tree SMOTE Regularized
- Random Forest Class Weighted Regularized
- Gradient Boosting Sample Weighted

The Random Forest Class Weighted Regularized model achieved the best overall macro F1-score of 0.3816. It also improved Fatal and Serious recall compared with the Majority Baseline model, which completely failed to detect minority classes.

Logistic Regression achieved the highest Fatal recall, but its overall macro F1-score was lower. This shows a trade-off between improving minority-class detection and maintaining balanced overall performance across all classes.

Random Forest performed best because ensemble tree models can learn complex non-linear relationships between road, environmental, and temporal features while remaining robust on noisy real-world tabular data.

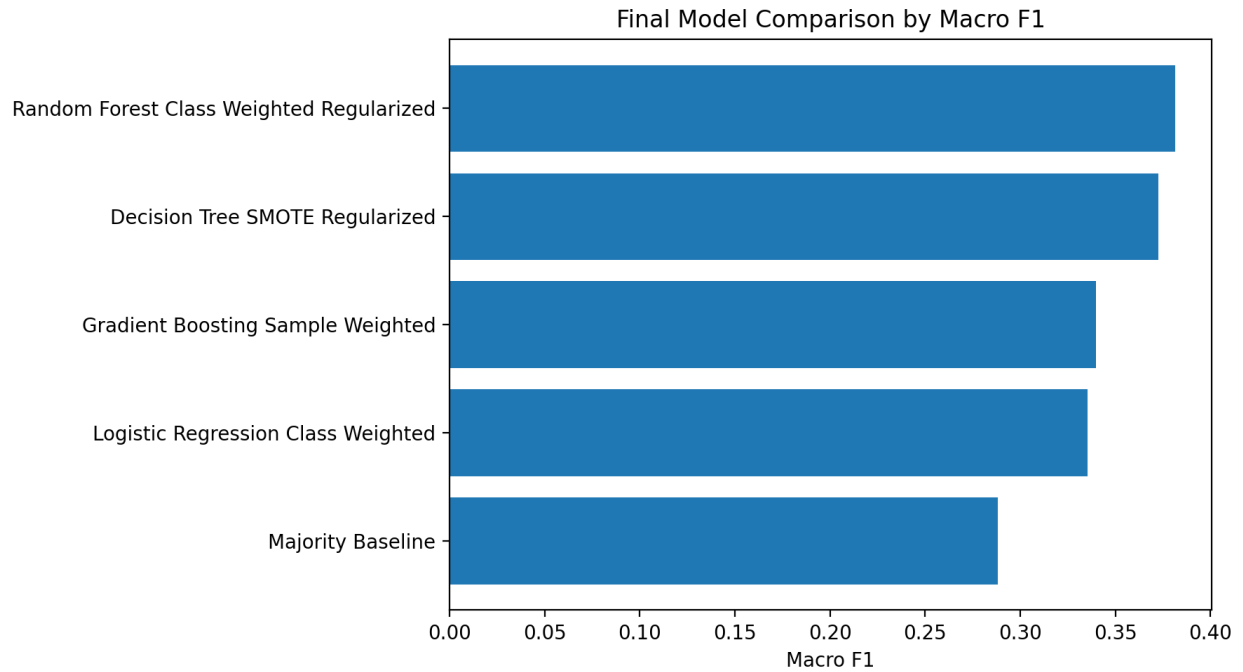


Figure 4. Final model comparison using macro F1-score.

Table 5. Final model comparison

Model	Accuracy	Macro F1	Weighted F1	Fatal Recall	Serious Recall	Slight Recall
Random Forest Class Weighted Regularized	0.5891	0.3816	0.6308	0.3059	0.3933	0.6524
Decision Tree SMOTE Regularized	0.6612	0.3727	0.6630	0.1645	0.2280	0.7987
Gradient Boosting Sample Weighted	0.4985	0.3399	0.5729	0.6151	0.2905	0.5578
Logistic Regression Class Weighted	0.4857	0.3355	0.5602	0.6217	0.3040	0.5368
Majority Baseline	0.7606	0.2880	0.6572	0.0000	0.0000	1.0000

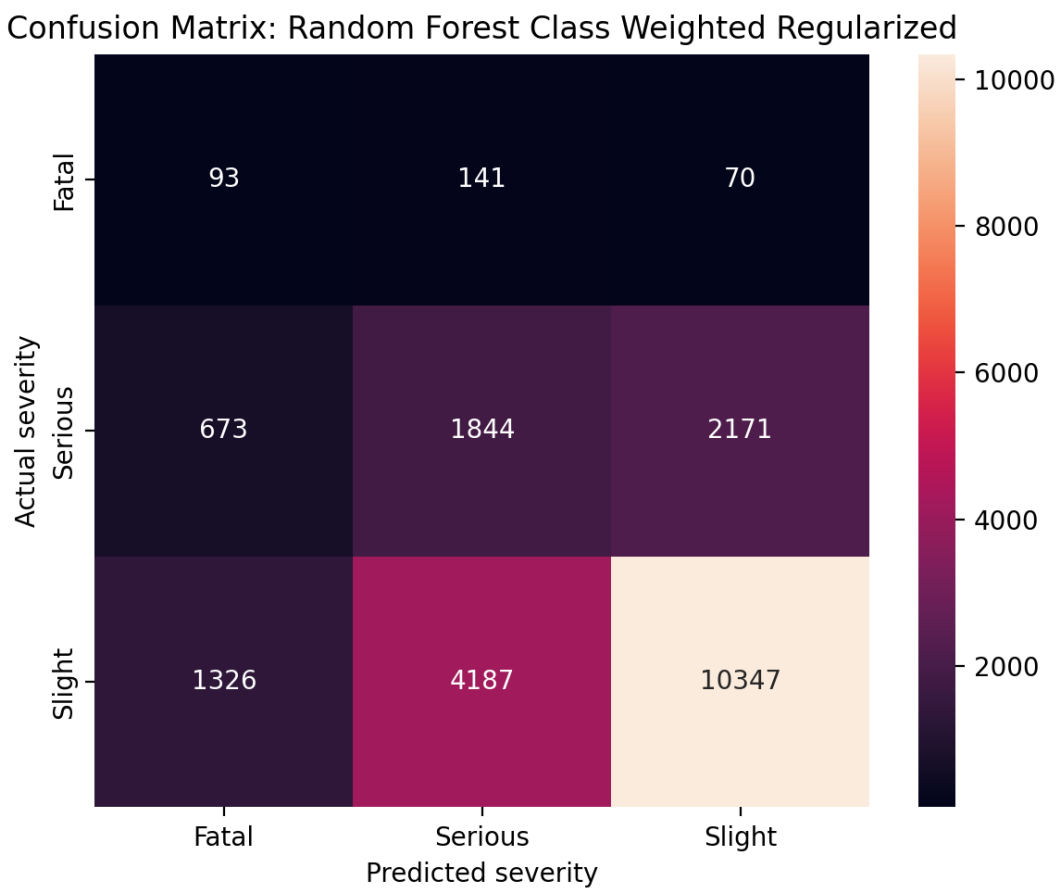


Figure 5. Confusion matrix of the best final model.

Table 6. Top 10 Random Forest feature importances

Feature	Importance
speed_limit	0.0997
hour	0.0769
month	0.0644
number_of_vehicles	0.0612
urban_or_rural_area_1	0.0431
urban_or_rural_area_2	0.0337
first_road_class_6	0.0208
light_conditions_1	0.0188
first_road_class_3	0.0182
road_type_6	0.0161

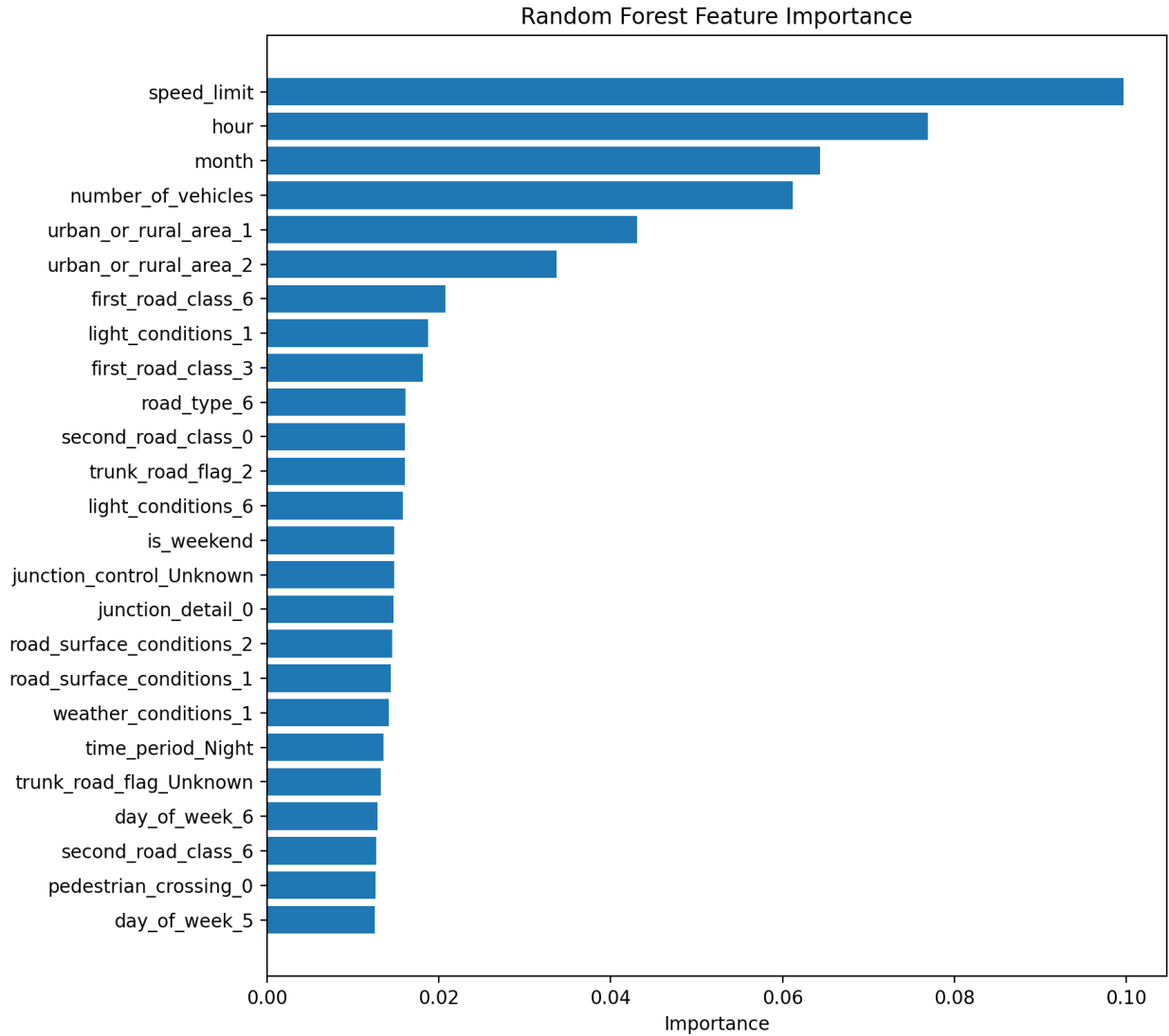


Figure 6. Top Random Forest feature importances.

4. Interpretation & Discussion

4.1 What the Imbalance Experiments Showed

The imbalance experiments showed that the original unbalanced training data was not sufficient for minority-class learning. The model trained without imbalance handling achieved 0.0000 Fatal recall and only 0.0591 Serious recall. This means the model mostly focused on the majority Slight class.

Different imbalance-handling methods produced different trade-offs. Class weighting improved Fatal recall to 0.3980 and Serious recall to 0.3752. Random oversampling improved macro F1-score to 0.3587, while SMOTE achieved the best controlled macro F1-score of 0.3727. Random undersampling achieved the highest Fatal recall but significantly reduced overall performance because too much majority-class information was removed.

These results show that handling class imbalance was necessary for responsible model evaluation and minority-class prediction.

4.2 Leakage Diagnostic and Solution

A diagnostic-only experiment intentionally included outcome-related columns such as:

- number_of_casualties
- enhanced_severity_collision
- collision_injury_based
- adjusted severity columns

The result was:

- 100% train accuracy
- 100% test accuracy
- 100% train macro F1
- 100% test macro F1

These results are unrealistic for a real prediction problem and clearly indicate data leakage. The model was effectively receiving information too close to the target label.

The solution was to remove all leakage-related columns from the final feature set before training the production models.

Table 7. Leakage diagnostic result

Experiment	Leakage Columns	Train Accuracy	Test Accuracy	Train Macro F1	Test Macro F1
Diagnostic model with leakage columns included	number_of_casualties, enhanced_severity_collision, collision_injury_based, collision_adjusted_severity_serious, collision_adjusted_severity_slight	1.0000	1.0000	1.0000	1.0000

4.3 Overfitting Diagnostic and Solution

Overfitting was evaluated by comparing train and test macro F1-scores. The largest overfitting gap was observed for the Random Forest Class Weighted Regularized model, with train macro $F1 = 0.4737$ and test macro $F1 = 0.3816$, giving a gap of 0.0921.

This indicates some overfitting risk but not extreme memorization. The model was controlled using:

- `max_depth`
- `min_samples_leaf`
- `max_features`
- class weighting
- regularization
- leakage removal

The Decision Tree model was also regularized to reduce overfitting risk.

Table 8. Overfitting diagnostics

Model	Train Macro F1	Test Macro F1	Macro F1 Gap	Train Weighted F1	Test Weighted F1	Weighted F1 Gap
Random Forest Class Weighted Regularized	0.4737	0.3816	0.0921	0.6842	0.6308	0.0534
Decision Tree SMOTE Regularized	0.4206	0.3727	0.0479	0.6828	0.6630	0.0197
Gradient Boosting Sample Weighted	0.3425	0.3399	0.0027	0.5725	0.5729	-0.0003
Majority Baseline	0.2880	0.2880	-0.0000	0.6572	0.6572	-0.0000
Logistic Regression Class Weighted	0.3347	0.3355	-0.0008	0.5601	0.5602	-0.0001

4.4 Practical Meaning of the Best Model

The best final model was Random Forest Class Weighted Regularized. Its macro F1-score was 0.3816, which was higher than the Majority Baseline and the other required models.

More importantly, the model improved Fatal and Serious recall compared with the baseline model, which completely failed to detect minority classes.

However, the overall performance remains limited because Fatal collisions are extremely rare and predicting collision severity using only road and environmental features is difficult. Therefore, the model should be viewed as a decision-support tool rather than a replacement for official collision investigation.

The feature importance analysis showed that speed limit, hour, month, number of vehicles, and urban or rural area were among the most influential variables in predicting collision severity.

4.5 Ethical and Practical Considerations

This project uses public, non-sensitive road safety data. The purpose is not to blame drivers or individuals, but to understand collision patterns that may support prevention planning and road safety improvement.

Because the dataset contains only reported collisions, the results may reflect reporting practices and should not be generalized beyond the dataset without caution.

The model should be used only as a supporting analytical tool and not as an automated decision-making system for legal or insurance purposes.

5. Conclusion

This project implemented a complete AI306 data mining workflow for predicting UK road collision severity using the 2023 UK Road Safety Collision dataset. Several important modelling problems were identified and solved during the project.

The dataset was highly imbalanced, with Fatal collisions representing only a very small percentage of the records. Because of this, multiple imbalance-handling methods were tested and compared using macro F1-score, minority recall, and confusion matrices instead of relying only on accuracy.

A leakage diagnostic experiment also showed that some outcome-related columns produced unrealistically perfect results, proving that they must be removed from the final models. Overfitting risk was evaluated using train-test metric gaps and controlled through model regularization and leakage prevention.

Among all tested models, the Random Forest Class Weighted Regularized model achieved the best overall balance with macro F1-score = 0.3816. The model improved Fatal and Serious recall compared with the Majority Baseline model, although predicting Fatal collisions remains difficult because the Fatal class is extremely rare.

Overall, the project demonstrates that responsible data mining requires not only building models, but also identifying and solving issues such as imbalance, leakage, misleading accuracy, and overfitting.

6. References

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3. [imbalanced-learn Documentation](#)
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Appendix A. Reproducibility Checklist

Random state was fixed to 42 for reproducibility.

The train-test split was stratified to preserve class distribution.

No leakage columns were used in the final production models.

Preprocessing pipelines were fit only on the training split.

No artificial row cap or sampling shortcut was used; the full cleaned dataset was used.

Multiple imbalance-handling methods were compared:

- No imbalance handling
- Class weighting
- Random undersampling
- Random oversampling
- SMOTE

Final models compared:

- Majority Baseline
- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting

Evaluation metrics included:

- Accuracy
- Macro F1-score
- Weighted F1-score
- Fatal recall
- Serious recall
- Confusion matrices

Leakage diagnostic experiments were performed and documented.

Overfitting diagnostics were performed using train-test metric gaps.

All charts, reports, confusion matrices, and CSV outputs were saved automatically to the outputs folders generated by the project code.

The project used Python with:

- pandas
- NumPy
- scikit-learn
- imbalanced-learn
- matplotlib
- seaborn

The final best model was:

- Random Forest Class Weighted Regularized

Final best macro F1-score:

- 0.3816